

CPR E 4890: Computer Networking and Data Communication

Lab Experiment #1 – Networking Utility Programs

(Total Points: 100)

Pre-lab Recommendations

Labs in this course will typically be conducted using the `4891labuser` account. This means you will not have access to your personal drive, and any files left on the lab computer will be visible to others. The TA recommends bringing a flash drive to easily bring home screenshots or files collected during the lab. Or, you are welcome to use any file transfer method you prefer, such as <https://www.file.io>.

Objective

To demonstrate how to use network diagnostic and probing tools such as `ping`, `nslookup`, `ifconfig`, `iperf`, `traceroute`, `tcptraceroute`, `Nmap`, `tcpdump`, `tcptrace`, and Wireshark.

Lab Expectations

Work through the lab individually and let the TA know if you have any questions. After the lab, write up a lab report and be sure to:

- Attend the lab. **(5 points)**
- Summarize what you learned in a few paragraphs. **(10 points)**
- Answer any questions asked in the exercises throughout the experiment. **(85 points in total, with 5 points for each of the 17 questions)**

Answers to questions shall contain:

- A screenshot including the command/action and its output
- A written comment or annotation of the screenshot (as appropriate)

Problem Description

In this lab experiment, you will learn about several network utility programs. Please use the “**Networking Commands Reference**” document provided alongside this lab description. For each set of exercises, read through the reference for the command and use manual pages if necessary.

Note: Network probing tools, such as `tcpdump` and `Wireshark`, are useful for analyzing network traffic and troubleshooting network problems. A number of privacy and security concerns are raised with the use of these tools – **please use them in an ethical manner.**

Warning: During this lab, do **not** log into any websites or remote applications. The tools we will be using allow anyone to see the content of packets on the wire. Hence, your username and/or password may be easy to discover.

Login Information

Make sure that you are logged in to your lab computer using the following credentials (otherwise, some commands that require “`sudo`” won’t work):

Username: `4891labuser`

Password: `4891labuser`

EXERCISES

ip a & ifconfig

1. What devices (interfaces) are available on your machine? What are the IP addresses assigned to them? (5 points)
2. What is the device (interface) named `lo` used for? Use manual pages or search online. What address is assigned to the device? (5 points)

ping

3. Ping the following destinations and report the average round-trip-time (the latency) to each one: (5 points)
 www.google.com
 www.iastate.edu
 vm.iastate.edu
 127.0.0.1
 www.cam.ac.uk
4. Rank the latency from least to greatest. Why do you suppose this is the case? Why are pings to 127.0.0.1 so much faster than pings to other destinations? Use the internet, manual pages, or ask your TA if stuck. (5 points)
5. Using the manual pages for ping (`man ping`), determine which flag/option can be used to specify the interface pings will be sent out of? Which option can be used to specify packet size? (5 points)

traceroute & tcptraceroute

6. Perform a traceroute to google.com. How many hops does it take for your packets to reach google.com? What is the latency? About how many hops are within the ISU network (example internal address: vm.iastate.edu)? (5 points)
7. Perform another traceroute to google.com using TCP packets instead of ICMP. Are there any differences in the output? (Hint: `sudo` is required for `tcptraceroute`) (5 points)

nslookup

8. Perform `nslookup` of www.iastate.edu. What is the IP address of this server? (5 points)
9. What is the mail exchanger for ece.iastate.edu? (5 points)
10. Which of the following are aliases? If it's an alias, record its canonical name. (5 points)
 www.apple.com
 apple.com
 www.google.com

iperf

11. Your TA is running the `iperf` server program on the machine 192.168.254.1. Run the `iperf` client program against the TA's `iperf` server and report the bandwidth of the connection in Gigabits per second (Gbps), Gigabytes per second (GBps), Megabits per second (Mbps), and Megabytes per second (MBps). (5 points)

nmap

12. Run a port map against your own machine. Is port 22 open? Which service runs on this port? (5 points)

tcpdump

13. For this exercise, work with a partner and alternate roles. Your machine is currently undergoing a ping flood attack! On partner #1's computer, execute ping [address] where "address" is partner #2's IP address. On partner #2's computer, use tcpdump and filter for ICMP packets to determine the IP address of the machine that is sending the packets. Include the IP addresses of your machine and your partner's machine in your report. (Note: use IP address 192.168.254.x where x can be found using earlier command; also, for a more accurate flood attack, -f can be used but requires sudo). (5 points)

Wireshark

Note: For each part of this exercise, make sure that previous attack simulations and other activities, such as browsing, are completed. This will provide a smoother experience and help you avoid dealing with too many unrelated packets during analysis.

To generate HTTP connections, open a web browser such as Firefox on the command line with, for instance, "firefox www.google.com". Let's analyze the TCP connections between your computer and www.iastate.edu.

Start a new capture and then connect to www.iastate.edu. Let it run for about 10 seconds. Use **Statistics-> Conversations -> TCP** to see all the TCP connections between your computer and various IP addresses. For this exercise, we are concerned with connections to the IP address corresponding to www.iastate.edu. (Hint: what command resolves a hostname to an IP address?). Sort the connections by **Address B** and take a **screenshot** of **only** the conversations between your IP address and the IP address of "www.iastate.edu". Then answer the following questions:

14. For **each** connection, indicate the duration and the number of packets sent from your computer during that connection. (5 points)
15. What were the destination ports used and what protocols do they correspond to? (5 points)

Continuing the tcpdump example, your computer is still under a ping flood (ICMP request and reply packets). Work with a partner and perform alternate roles. Start a new capture, and let it run for about 10 seconds.

16. Determine how much data (in bytes) each ICMP packet contains. (Hint: the conversations panel does not support ICMP explicitly; you can capture the traffic between your two nodes to make sure the result is valid) (5 points)

Start a new capture, and let it run while you complete a traceroute and tcptraceroute to www.ebay.com.

17. What types of packets are sent with traceroute? What types of packets are sent with tcptraceroute? (5 points)